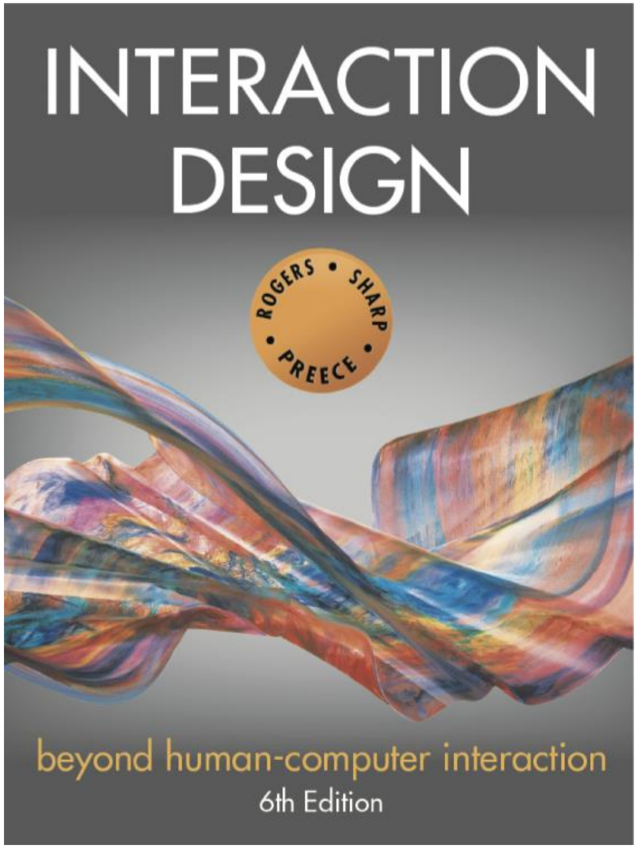
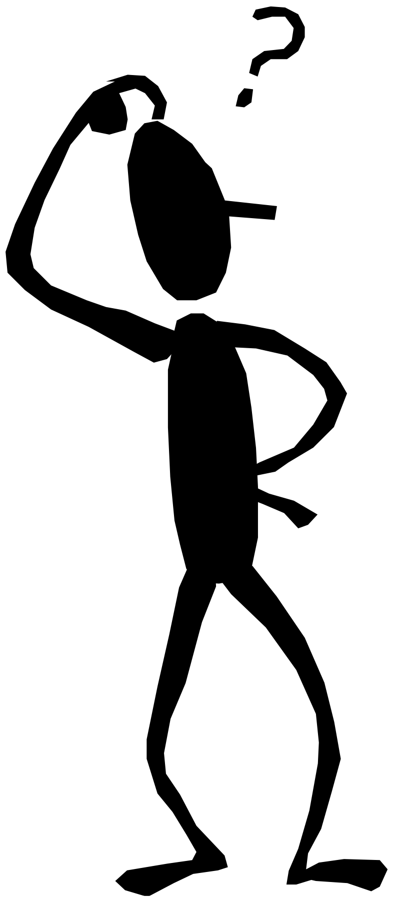


Chapter 12

DESIGN, PROTOTYPING and CONSTRUCTION

Overview

- Prototyping
- Conceptual design
- Concrete design
- Generating prototypes
- Construction



Designing with or for People?

- Is stakeholder engagement one-way?
- Co-design
  - Emphasizes creativity and mutual learning
  - Often multidisciplinary
- Participatory design (Scandinavian version)
  - Stakeholders are design partners
  - Co-operative prototyping is important
- Community-based design
  - Scale in terms of numbers and diversity

Prototyping

- What is a prototype?
- Why prototype?
- Different kinds of prototyping
  - Low fidelity
  - High fidelity
- Compromises in prototyping
  - Vertical
  - Horizontal
- Final product needs to be engineered

What is a prototype?

- One manifestation of a design that allows stakeholders to interact with it
- In other design fields, a prototype is a small-scale model:
  - A miniature car
  - A miniature building or town

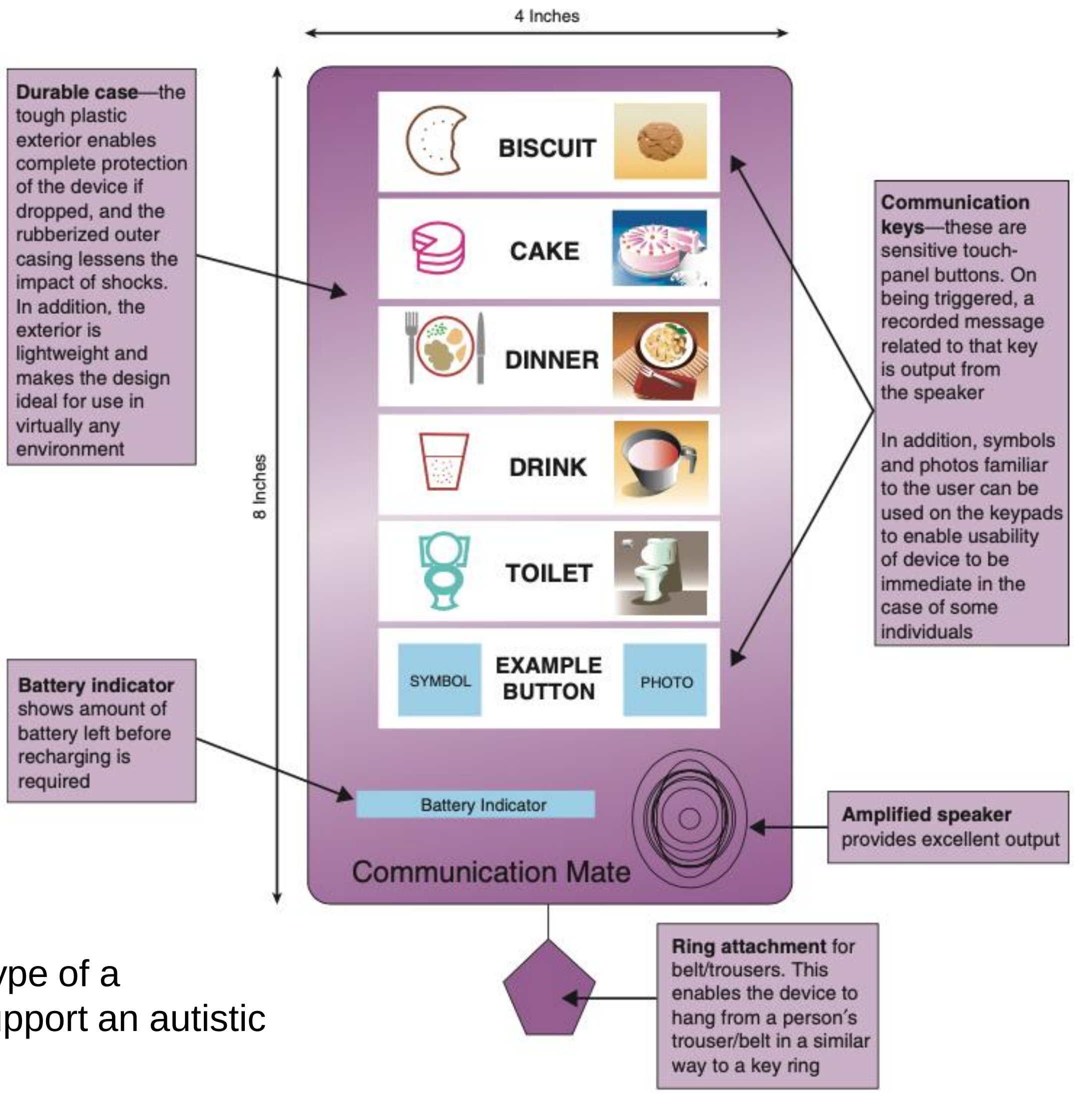
Source: [PalmPilot wooden model](#)
© Mark Richards

What is a prototype in interaction design?

Among other things an interaction design prototype can be:

- A series of screen sketches
- A storyboard, for example, a cartoon-like series of scenes
- A PowerPoint slide show
- A video simulating the use of a system
- A lump of wood (for instance, the PalmPilot)
- A loosely connected set of electronic elements
- An animation of product use
- A piece of software with limited functionality written in the target language or in another language

Example interaction design prototype



A paper-based prototype of a handheld device to support an autistic child
Used courtesy of Sigil Khwaja

Why prototype?

- Evaluation and feedback are central to interaction design
- Stakeholders can see, hold, and interact with a prototype more easily than a document or a drawing
- Team members can communicate effectively
- Ideas can be tested out
- Prototyping encourages reflection: an important aspect of design
- Prototypes answer questions and support designers in choosing between alternatives

Low-fidelity Prototyping

- Uses a medium which is unlike the final medium, for example, paper or cardboard
- Is simple, quick, cheap & easily changed
- Examples:
    - Sketches of screens and task sequences
    - Index cards or sticky notes
    - Storyboards
    - Wizard-of-Oz

Storyboards

- A series of sketches showing how someone might progress through a task using the product

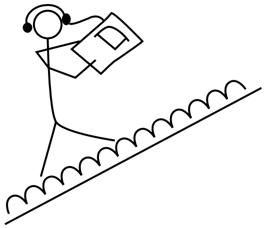
- Often used with scenarios, bringing in more detail and a chance to role play

- how a **user interacts with a system step-by-step** through a sequence of sketches or panels

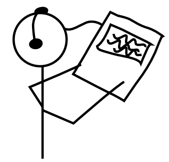
Storyboards

- Each panel typically shows:
  - The user
  - The context or environment
  - The interface interaction
  - The result or system response

Example storyboard

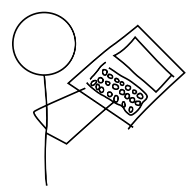


Christina walks up hill; the product gives her information about the site

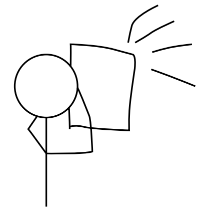


Christina adjusts the preferences to find information about the pottery trade in Ancient Greece

Christina scrambles to the highest point

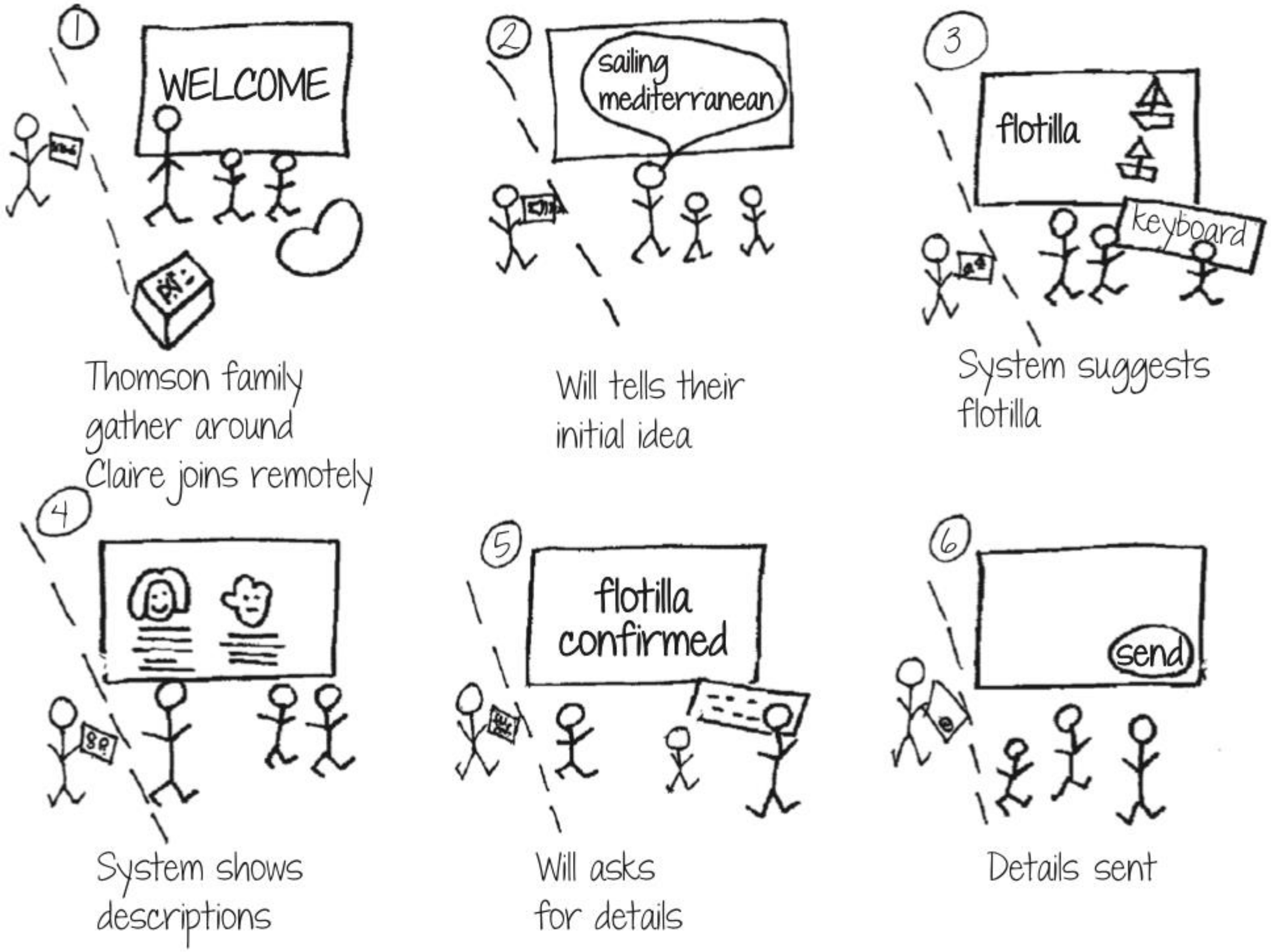


Christina stores information about the pottery trader's way of life in Ancient Greece



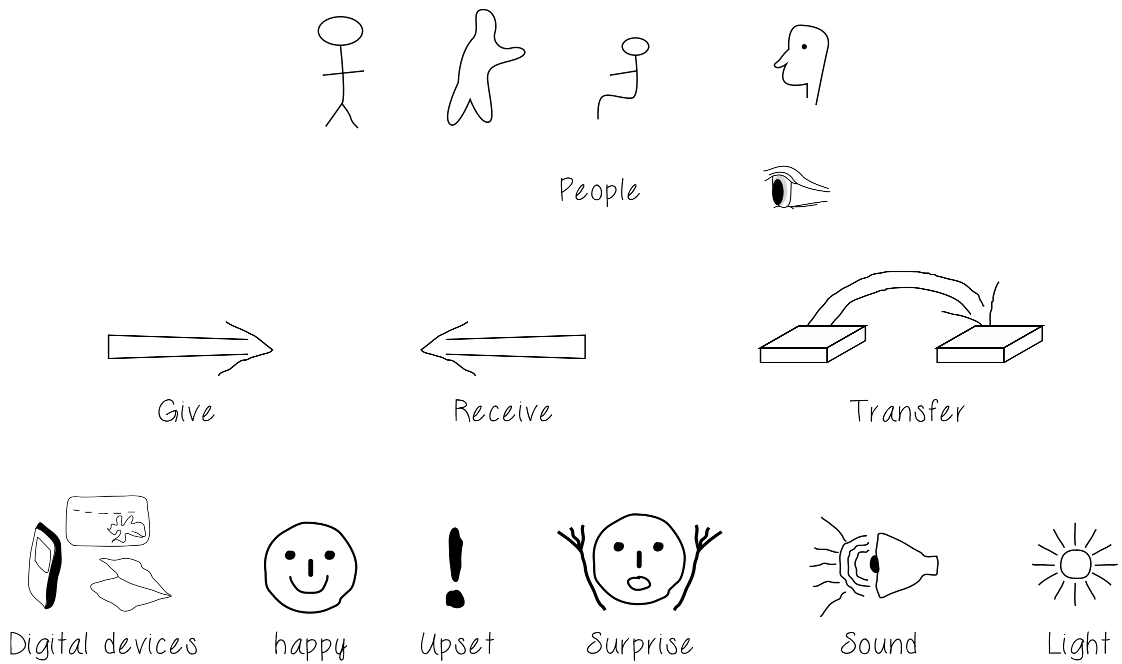
Christina takes a photograph of the location of the pottery market

Another Example

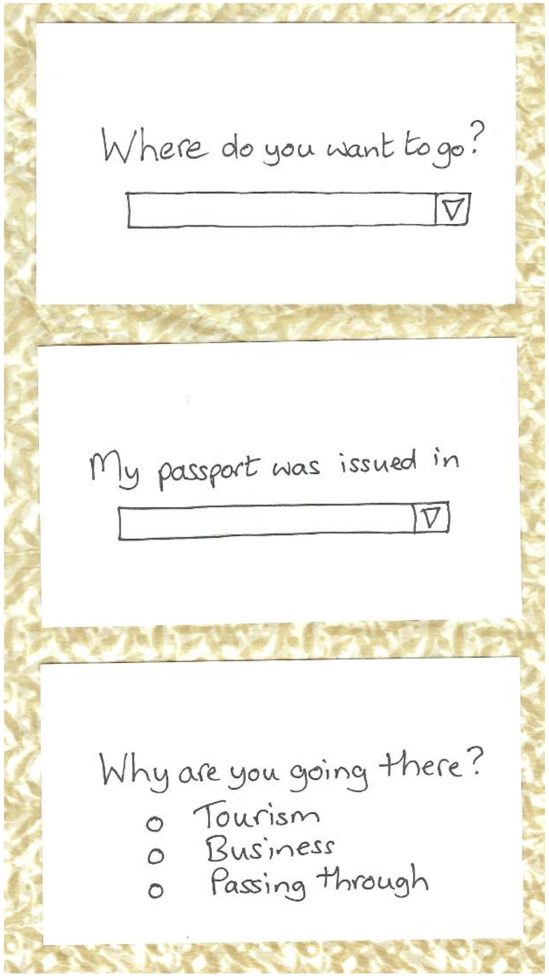


Sketching

- Low-fidelity prototyping often relies on sketching
- Don’t be inhibited about drawing ability — Practice simple symbols



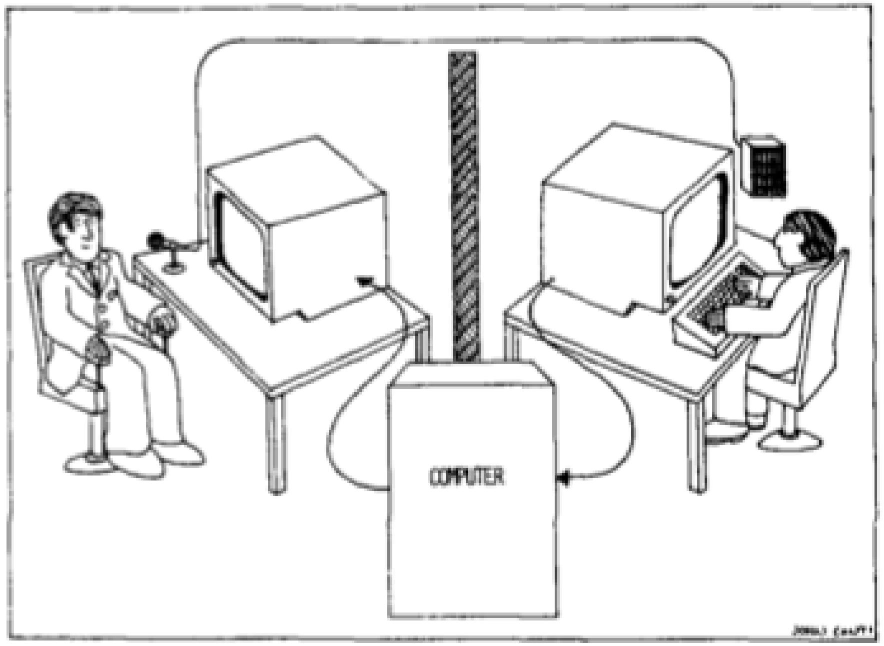
Prototyping with index cards



- Index cards (3 x 5 inches)
- Each card represents one element of interaction
- In evaluation, can step through the cards

Wizard-of-Oz prototyping

- The participant thinks they are interacting with a computer, but a human is responding to output rather than the system
- Usually done early in design to understand peoples' expectations



- What ethical issues are there with this approach?

High-fidelity prototyping

- Uses materials that would be in the final product
- Prototype looks more like the final system than a low-fidelity version
- High-fidelity prototypes can be developed by integrating existing hardware and software components
- Danger that people think they have a complete system…see compromises

Compromises in prototyping

- Prototyping involve compromises
- For software-based prototyping, maybe there is a slow response? sketchy icons? limited functionality?
- In-the-wild prototypes are operational but not necessarily robust
- Two common types of compromise:

  Horizontal: wide range of functions, with little detail

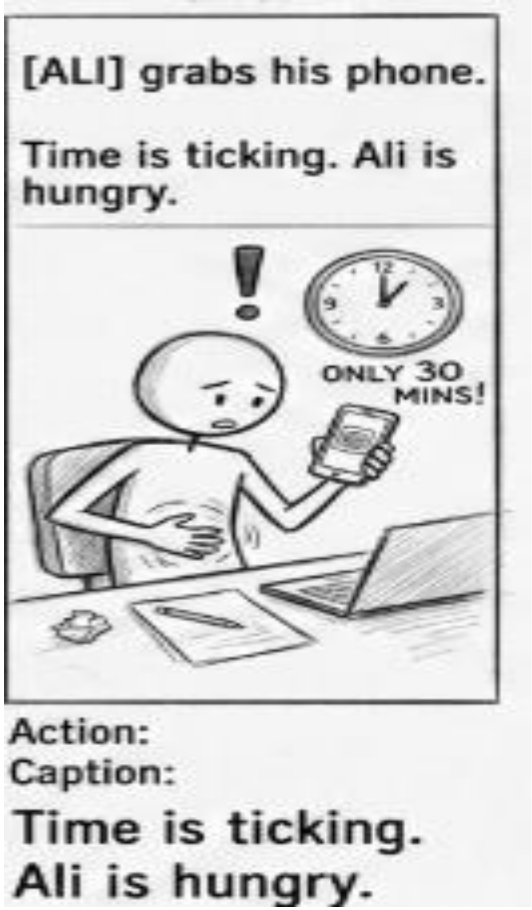
  Vertical: a lot of detail for only a few functions
- Other common compromise: robust vs changeable
- Beware of compromises in prototypes as final product needs to be engineered

Class Activity

- Design a storyboard for following scenario
- Ali has a short break and wants to order food using a food delivery app.
  - Ali has a 30-minute break and feels hungry
  - Ali opens the food delivery app
  - He browses nearby restaurants
  - Ali selects a meal and places the order
  - Order confirmed. Delivery in 20 minutes

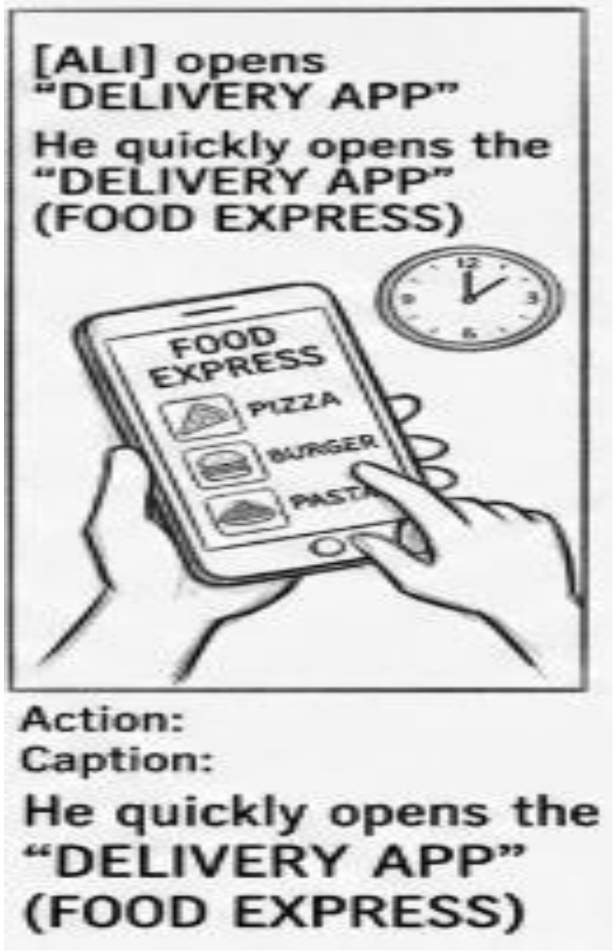
Panel 1

- Stick figure of Ali
- Clock showing limited time.
- Ali holding phone
- Ali feels hungry



Panel 2

- Phone screen
- App icon
- Ali tapping the screen



Panel 3

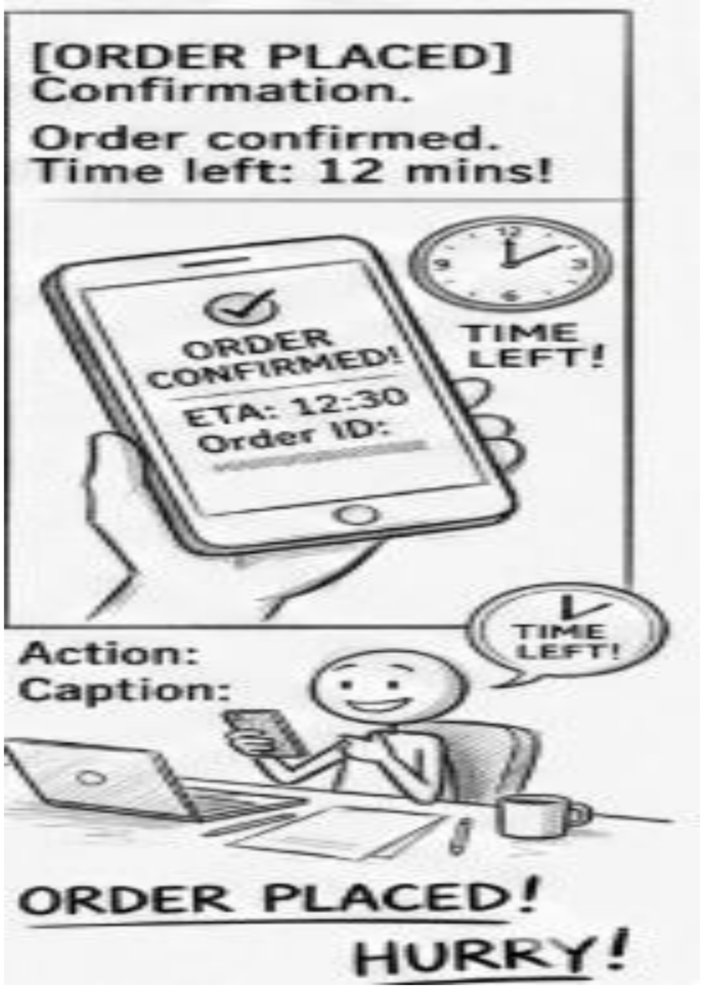
- Phone screen with list of restaurants
- Ali scrolling

Panel 4

- Phone screen with food item
- Button labeled "Order"

Panel 5

- Order confirmed
- Delivering in 20 min



Conceptual design

- A conceptual model is an outline of what people can do with a product and what concepts are needed to understand how to interact with it
- Creativity and brainstorming techniques
- Consider alternatives: scenarios and prototyping help

Interface metaphor

- An **interface metaphor** is a design concept where **elements of a user interface are based on familiar real-world objects or experiences**
- helping users understand how the system works.
- It uses **users' existing knowledge** to make the interface intuitive

Choosing an interface metaphor

- Interface metaphors combine familiar knowledge with new knowledge in a way that will help the user understand the product.
- Three steps: understand functionality, identify potential problem areas, and generate metaphors
- Evaluate metaphors:
  - How much structure does it provide?
  - How much is relevant to the problem?
  - Is it easy to represent?
  - Will the audience understand it?
  - How extensible is it?

Examples of Interface Metaphors

Interface Feature	**Real-World Metaphor**
Desktop interface	Office desk
Folder icons	Physical folders
Trash / recycle bin	Waste basket
Shopping cart in e-commerce	Real shopping cart
Calendar apps	Paper calendar

Wireframes

- A **wireframe** is a **low-fidelity visual representation of a user interface layout** that shows:
  - Screen structure
  - Placement of interface elements
  - Navigation components
  - Information hierarchy
- **do not focus on colors, graphics, or visual design**, but on **structure and functionality**

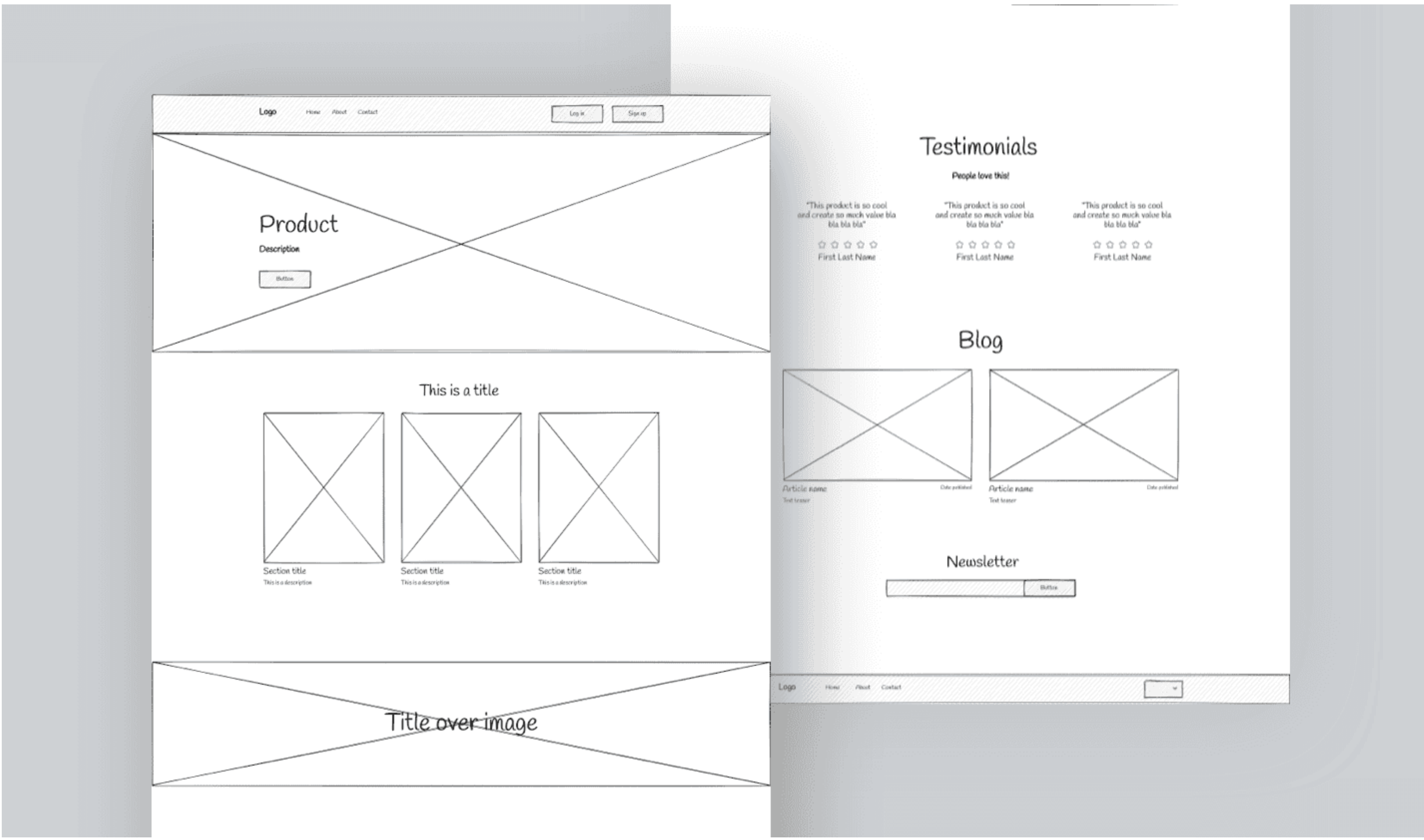
Wireframes

- **What wireframes typically show**
- Buttons
- Menus
- Navigation bars
- Input fields
- Content areas
- Labels or placeholders

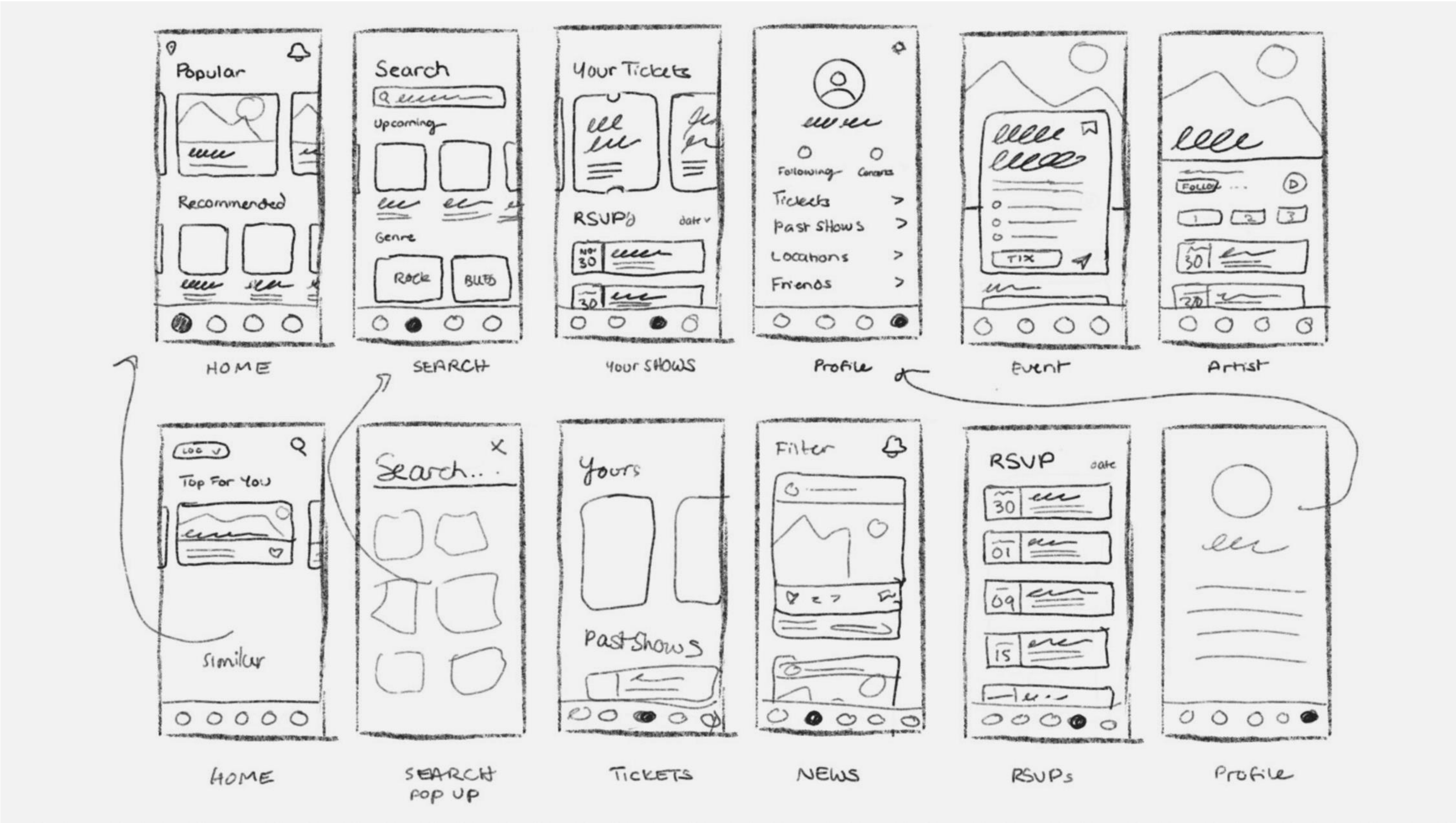
Wireframes

- **The "Rules" of Wireframing:**
- **No Color:** Use only grayscale (white, black, and gray).
- **Placeholders:** Use an "X" in a box to represent an image and "Lorem Ipsum" or simple lines for text.
- **Standard Symbols:** Use common UI patterns (magnifying glass for search, three lines for a menu).

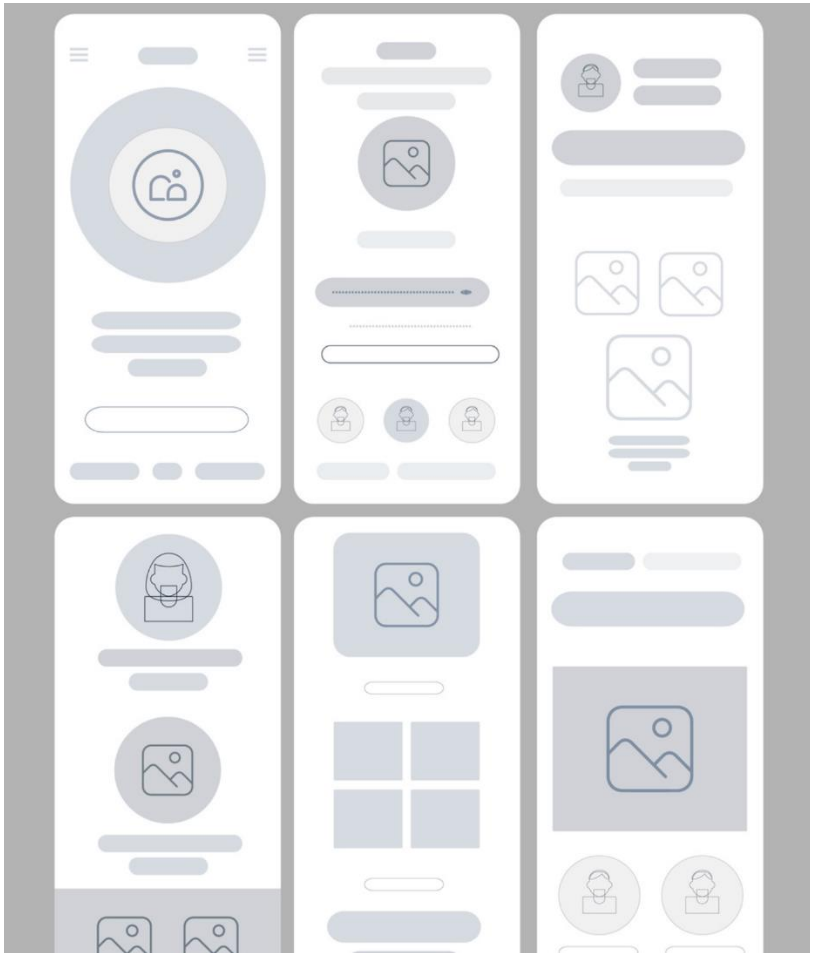
Wireframes Examples



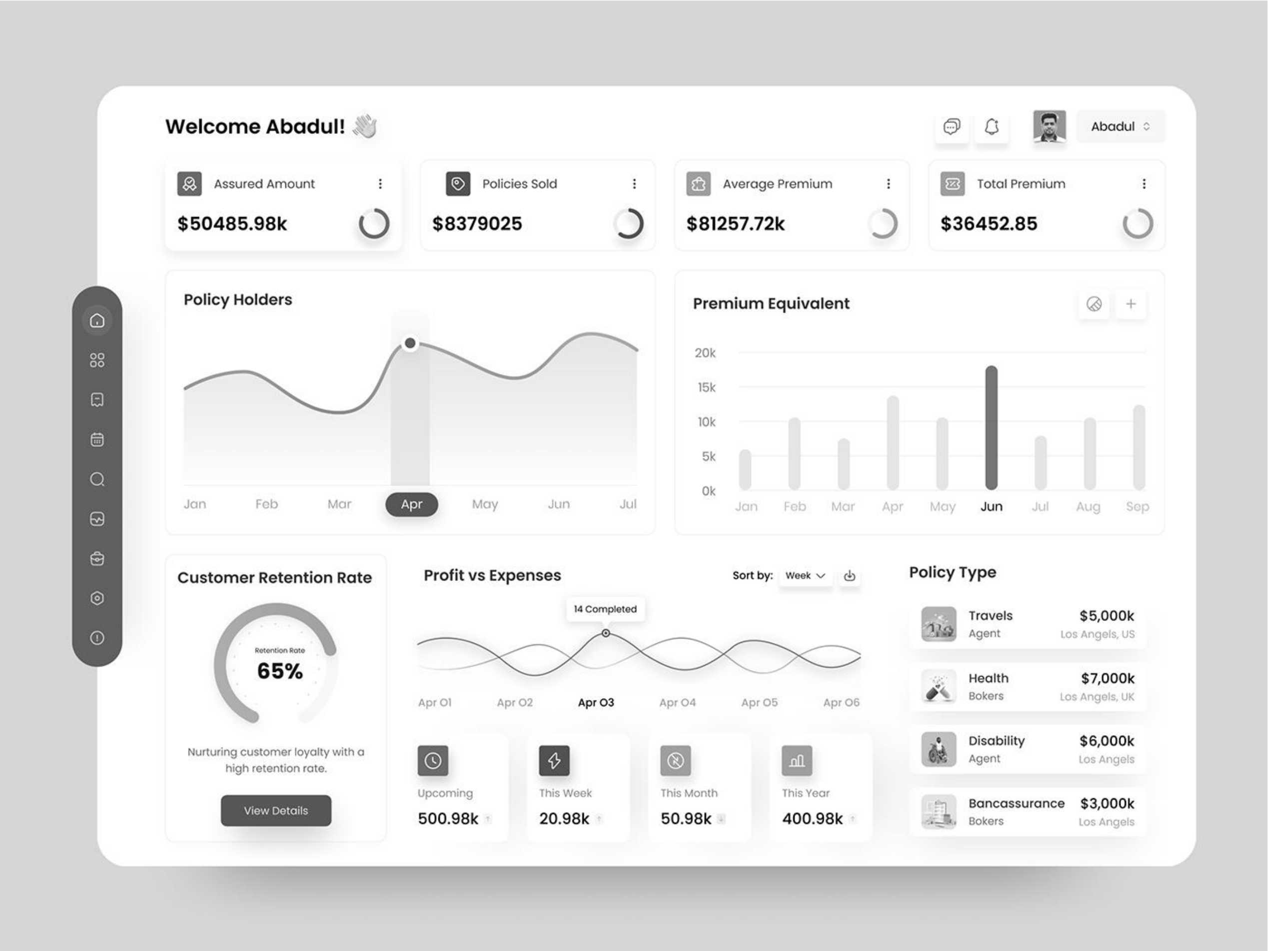
Wireframes Examples



Wireframes(using design tools)



Wireframes dashboard example



Why Wireframes are Important in User-Centered Design

- Translate scenarios and storyboards into interface screens
- Focus on layout and usability rather than aesthetics
- Identify navigation problems early
- Enable fast design iteration
- Facilitate communication with developers and stakeholders

Class Activity

Ali is in a rush and wants to order his "usual" burger.

The Goal: Design a wireframe for the **Home Screen** of the food app that prioritizes speed for a user like Ali.

- **Top Section:** Search bar and Delivery Address.
- **Middle Section:** "Order Again" horizontal scroll (so Ali can find his burger in one tap).
- **Bottom Section:** Categories (Pizza, Burgers, Sushi) using simple icons.
- **Footer:** Navigation bar (Home, Search, Orders, Profile).

Type	Advantages	Disadvantages
Low-fidelity prototype	• Quick revision possible • More time can be spent on improving the design before starting development • Evaluates multiple design concepts • Useful communication device • Proof of concept	• Limited error checking • Poor detailed specification for development • Facilitator-driven • Limited usefulness for usability tests • Navigational and flow limitations
High-fidelity prototype	• (Almost) complete functionality • Fully interactive • User-driven • Clearly defines navigational scheme • Use for exploration and test • Look and feel of intended product • Serves as a "living" or evolving specification • Marketing and sales tool	• More resource-intensive to develop • Time-consuming to modify • Inefficient for proof-of-concept designs • Potential of being mistaken for the final product • Potential of setting inappropriate expectations

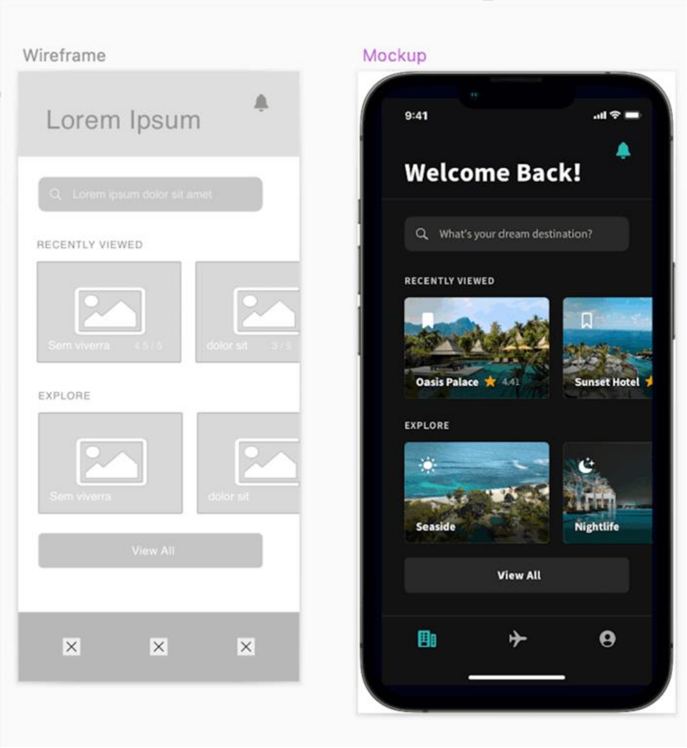
Table 12.3 Advantages and disadvantages of low- and high-fidelity prototypes

Lo-fi Vs. Hi-fi example

Wireframe (le

Vs.

Mockup (right



Summary

- Prototyping may be low fidelity (such as paper-based) or high fidelity (such as software-based)
- Ready-made software and hardware helps create prototypes
- Two aspects to design: conceptual and concrete
- Conceptual design develops an outline of what people can do and what concepts are needed to understand the product.
- Concrete design specifies design details, for example, layout or navigation

Summary (*continued*)

- Three approaches to develop an initial conceptual model: interface metaphors, interaction styles, and interface styles.
- Expand an initial conceptual model by considering whether product or user performs each function, how those functions are related, and what information is required to support them
- Scenarios and prototypes can be used to explore design ideas
- Physical computing kits and software development kits facilitate the transition from design to construction